

Report:

The "Simple" Explanation

Imagine a huge mall where thousands of people walk in every day.

- **The Problem:** We noticed that in the electronics section, people are picking up expensive items (like phones or laptops) and putting them in their shopping carts, but then they just **leave the cart in the middle of the aisle** and walk out of the mall without paying.
 - **Our Job:** We are like "Data Detectives." We are going to look at the security logs (the dataset) to see if they left because the price was too high, because they didn't trust the brand, or because it was too late at night and they got tired.
 - **The Result:** We will tell the mall manager exactly how to stop people from leaving those carts behind so the mall can make more money.
-

3-Week Roadmap & Task Distribution

Since you are in your 2nd year and this is your first time using **Pandas** and **NumPy**, we will work in a "linear" way. This means everyone starts together, then branches out.

Week 1: Data Survival & Cleaning

Goal: Get the "Security Logs" ready to read.

- **Student 1 (Lead Technician):** Install Python/Jupyter. Load the dataset and filter it down. (The file is huge, so you'll only keep the "Electronics" rows so it doesn't crash your computers).
- **Student 2 (Data Cleaner):** Find where the "Brand" or "Category" names are missing and fill them in with "Unknown" so the code doesn't break later.

Week 2: The "Detective" Investigation (EDA)

Goal: Find the patterns of why people leave.

- **Student 3 (Conversion Analyst):** Calculate the "Funnel." Out of 100 people who viewed a product, how many put it in a cart? How many actually bought it?
- **Student 4 (Price & Brand Specialist):** Group the items by price (e.g., \$0-100, \$100-500, \$500+). Check which group has the most "left-behind" carts.

Week 3: The "Evidence" & Presentation

Goal: Show the results and suggest solutions.

- **Student 5 (Visualization Lead):** Take the numbers from Students 3 and 4 and turn them into clear charts (Bar charts for brands, Line charts for time of day).
- **All Students:** Together, you will write the 3 recommendations (like "Give a discount at 9 PM since that's when people leave carts").

MarketPulse AI: Project File Structure

Here is how your project folder should look, and who is responsible for each part:

1. **main_dataset.csv**

- **Indicates:** This is the "Raw Data" (the huge file from Kaggle).
- **Owner:** Everyone starts here.

2. **1_data_cleaning.ipynb**

- **Indicates:** The script that removes errors and filters the data to just "Electronics."

- **Owner: Student 1 (Lead Tech) & Student 2 (Data Cleaner).**
- **Status: CRITICAL BLOCKER.** If this file isn't finished, no one else can start because the data will be too "messy" or too large to process.

3. **2_behavior_analysis.ipynb**

- **Indicates:** The math behind the "Purchase Funnel" (Views $\rightarrow$ Carts $\rightarrow$ Purchases).
- **Owner: Student 3 (Conversion Analyst).**
- **Dependency:** Blocks Student 5. You cannot visualize the funnel until the math in this file is done.

4. **3_market_research.ipynb**

- **Indicates:** The analysis of Price vs. Brands.
- **Owner: Student 4 (Market Researcher).**
- **Dependency:** Blocks Student 5. You cannot recommend price changes until this analysis is finished.

5. **4_final_dashboard.ipynb (or Power BI file)**

- **Indicates:** The final visual charts and the 3 business recommendations.
- **Owner: Student 5 (Visuals Lead).**
- **Dependency:** This is the "End Point." It is blocked by everyone else.

6. **final_report.pdf**

- **Indicates:** The written answers to your Dr.'s questions.
 - **Owner: All 5 Students** (Contribution from each role).
-

The "Blocker" Map (Who is waiting for whom?)

To finish in **3 weeks**, you must respect these "Stop Signs":

- **The Ultimate Blocker (Week 1):** Student 1 and 2 **block everyone**. If the cleaned dataset isn't ready by the end of Week 1, the Analysts (Students 3 and 4) will have nothing to work on in Week 2.
- **The Middle Blockers (Week 2):** Students 3 and 4 **block Student 5**. Student 5 cannot build a dashboard with "empty" numbers. They need the tables and results produced in notebooks #2 and #3.
- **The Final Polish (Week 3):** All files block the **Final Report**. You cannot write your "Evidence" until the code and charts are 100% complete.